

Travis Mendoza

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PROFILE

Systems and software engineer bridging hardware, simulated environments, and ML infrastructure. Proven ability to build end-to-end data pipelines for physical sensor arrays, develop complex systems architectures, and prototype in C++, Python, and ROS 2. Deep experience operating at the intersection of embedded hardware, autonomous systems, and advanced robotics (IoT networks, cislunar navigation architecture, and mechatronic systems). Bringing technical rigor, rapid-iteration 0→1 execution, and systems-level modeling capabilities to generative modeling and physical intelligence research.

EXPERIENCE

Founding Engineer | EyeZense (Stealth Startup)

October 2024 – January 2026

- Led end-to-end development of IoT robotics systems, building infrastructure to capture, stream, and annotate physical world data (video, audio, chemical) for machine learning model training.
- Implemented perception and autonomy workflows in C++, Python, and ROS 2; managed real-time object and sound detection streams from custom sensor arrays and mobile robots.
- Engineered data collection pipelines specifically designed to capture proprietary nano chemical sensor data, establishing the foundation for bespoke ML inference on hardware edge devices.
- Defined system-level interfaces, managing technical integration across hardware (sensors, compute) and software subsystems (V&V, autonomy, edge inference).
- Served as primary technical communicator during fundraising, orchestrating live technical demonstrations that secured a \$15M Seed round.

Customer-Facing Test Engineer | Skyloom

June 2022 – March 2024

- Owned the verification and validation (V&V) pipeline for highly precise mechatronic optical communication systems (space-based laser comms), translating strict requirements into formal test matrices.
- Developed data collection and analysis pipelines to process and interpret mechanical, electrical, and optical test data, driving iterative design optimizations across engineering teams.
- Authored the critical Acceptance Procedure for the mechanical Fast Steering Mirror (FSM), standardizing the evaluation criteria for the core optical pointing and tracking hardware in a high-volume manufacturing environment.

Technical Staff Intern | The Aerospace Corporation

June 2020 – August 2021

- Co-designed formal system architectures for cislunar Position, Navigation, and Timing (PNT) infrastructure as part of a US Space Force-funded Concept Design Center study.
- Developed analysis software in MATLAB to model and compute Lie group and Lie algebra transformations for spacecraft attitude determination and control systems (ADACS).

Mechanical Engineering Intern | NASA Ames Research Center

May 2013 – August 2013 & May 2015 – August 2015

- Designed, prototyped, and tested the physical docking interface components for Astrobee (ISS autonomous free-flying robot), enabling unassisted recharging and docking in microgravity.
- Operated multi-variable propulsion tests in LabVIEW and processed experimental data via MATLAB/spreadsheets to determine the dynamic impact of varying impeller volumes and nozzle types.

EDUCATION

M.S., Mechanical Engineering (Controls Emphasis) | University of California, Berkeley | 2020–2021

- National GEM Consortium Fellow.
- Thesis: "Design Concepts for a Position, Navigation, and Timing Service for the Earth-Moon Corridor" — evaluated limitations of GNSS for cislunar spacecraft through simulation and data analysis.
- Research Assistant (MPC Lab): Documented setup procedures and infrastructure for CARLA, an open-source simulator for autonomous driving research on Ubuntu.

B.S., Mechanical Engineering | University of Southern California | 2012–2016

TECHNICAL SKILLS

Software & Infrastructure: Python, C++, ROS 2, MATLAB, Django, SwiftUI.

Data & ML Analytics: NumPy, Pandas, Matplotlib, Data Annotation Workflows, LLM API Integration.

Hardware & Robotics: Sensor Data Pipelines, Embedded Systems, Real-Time Inference, Edge Compute.

Systems Engineering: Systems Architecture Modeling, Physical World Simulators (CARLA), V&V, Fast-Iterative Prototyping.